PHAS0056 Mini Project: Jet Classification at the LHC

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Abstract

Accurate classification of particle jets is essential for interpreting the large volume of collision data produced at the Large Hadron Collider. This work presents a systematic comparison of five deep-learning architectures for tagging jets as being initiated by gluon, light quark, W, Z, or top; the models exploit multiple input modalities, including jet images, ECAL and HCAL calorimeter views, and high-level kinematic features, combined through convolutional and fully connected neural networks. The best model, a multimodal CNN–MLP using all image representations and 16 features, achieves a test accuracy of 97.8% and F_1 score of 0.976, showing that combining image-based and scalar features provides the strongest classification performance.

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1 Introduction

The Large Hadron Collider (LHC) provides the highest-energy proton–proton collisions currently accessible in the laboratory and is therefore a unique environment for testing the Standard Model and searching for new physics. A large fraction of these collisions produce quarks and gluons, which cannot be observed directly due to colour confinement, but instead fragment and hadronise into collimated sprays of particles known as jets; as a result jets are among the most common and phenomenologically important objects reconstructed in collider experiments, and their detailed analysis is central to a broad range of measurements at the LHC [1].

1.1 LHC Jet Production

Jet physics becomes especially rich in the boosted regime: when heavy particles such as the W and Z bosons, the Higgs boson, or the top quark are produced with large transverse momentum, their hadronic decay products may become sufficiently collimated to be reconstructed as a single jet, and in such cases, the internal energy flow and radiation pattern of the jet contain valuable information about its origin. The study of these internal properties, known as jet substructure, provides a powerful way to distinguish ordinary QCD jets from jets arising from heavy-particle decays [2].
More recently, machine learning has become increasingly important in this field: a notable development was the introduction of jet images, in which calorimeter energy deposits are represented as pixels in an image and analysed using computer-vision-inspired techniques. This approach showed that low-level detector information could be used directly for jet tagging, often providing additional discriminating power beyond traditional handcrafted observables [3]. Modern deep learning methods have pushed this idea further by operating on particle-level representations of jets, with graph neural networks and transformers, like Particle Transformer, capturing constituent interactions [2].
In this report, jet tagging is formulated as a multiclass classification problem in which reconstructed jets are assigned to one of five categories: gluon, light quark, W boson, Z boson, or top quark. Different neural-network architectures are investigated using image-based and scalar kinematic representations of the jets, with the aim of assessing how the choice of input representation and model design affects classification performance.

1.2 Neural Networks

Neural Networks (NNs) are machine learning models composed of interconnected computational units, or neurons, arranged in input, hidden, and output layers; the simplest type of neural networks are feed-forward NNs, in which each neuron receives a set of inputs, applies weights and a bias term, and then passes the result through an activation function before transmitting it to the next layer. Activation functions are essential because they introduce non-linearity into the network, allowing the model to learn complex mappings between inputs and outputs that cannot be represented by purely linear transformations; common examples include the ReLU, sigmoid, and softmax functions, each of which is suited to different stages of the learning process and different output requirements [4].
The performance of a neural network depends not only on its architecture, but also on the way it is trained and evaluated, e.g. in classification tasks, the final output layer is often chosen to match the structure of the problem, whereas for multiclass classification, a softmax activation is typically used to convert the network outputs into a probability distribution over the possible classes. Model performance can then be assessed using metrics such as accuracy (which in our case is the ability of

the network to identify the origin of a jet based on data inputted in it) , loss, precision, recall, and F1 score, depending on the balance and objectives of the specific dataset [5].

Convolutional Neural Networks (CNNs) are a specialised class of feed-forward neural networks designed for grid-like data such as images: instead of connecting every neuron to every input value, CNNs use convolutional filters that slide across the input and detect local spatial patterns such as edges, shapes, and textures. These operations produce feature maps, which are commonly followed by pooling layers that reduce dimensionality while preserving the most informative features (as shown in Fig. 1); furthermore, by stacking multiple convolutional and pooling layers, CNNs can learn increasingly abstract representations of the input data, making them particularly effective for image classification problems [6].

Modern CNN architectures also commonly include regularisation techniques to improve generalisation and reduce overfitting. One widely used method is dropout, in which a random subset of neurons is temporarily removed during training so that the network does not become overly dependent on specific activations or connections, this improves robustness and often leads to better performance on unseen data. In this project, CNN-based models are well suited to the jet-tagging task because jet images can be treated as image-like inputs, allowing the network to learn discriminative spatial features directly from the detector-level representations; furthermore, to complement the standard CNN, a multi-input architecture that uses both image representations and jet level features was implemented , enabling us to evaluate how the inclusion of additional feature information affects model accuracy and training efficiency.

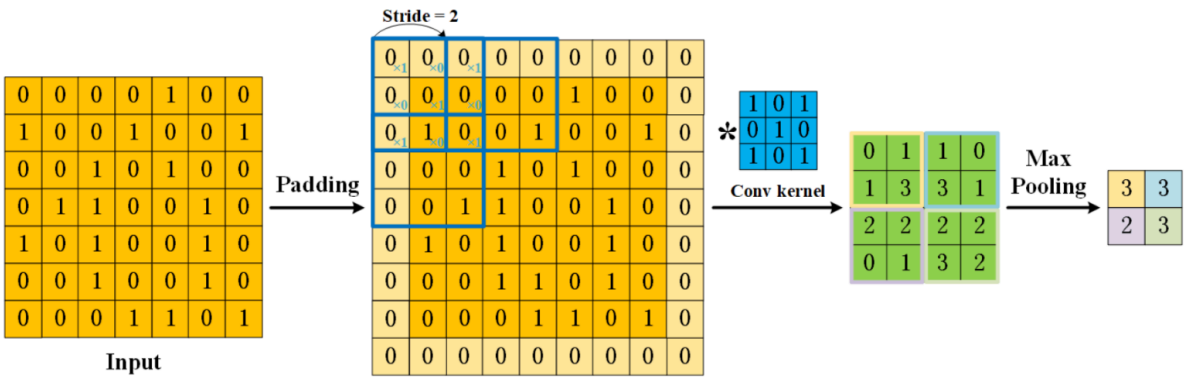

Figure 1: *Illustration of the main operations in a 2D convolutional neural network. The input image is first padded, then convolved with a kernel to generate a feature map, and finally downsampled using max pooling to retain the most important features while reducing dimensionality.*

2 Methodology

2.1 Data Preprocessing

The dataset used in this project is the publicly available HLS4ML LHC Jet dataset (100 particles), released for FastML/HLS4ML studies and available on Zenodo [7] and consisting of simulated high-p_T jets produced in proton–proton collisions at the LHC. Each jet belongs to one of five classes according to its origin: gluon (g), light quark (q), W boson, Z boson, or top quark (t); additionally the dataset provides several complementary representations of each jet, including image-based detector views and high-level engineered features, making it well suited to the study of both single-input and multimodal

neural network architectures.

Three jet representations were used in this work: the first one is a 100×100 jet image, which may be interpreted as a pixelised map of jet constituents in in the (η, ϕ) plane and is constructed from the highest-p_T constituents of the jet (as can be seen in Fig. 2); the second one is a set of high-level features (HLFs), which summarise physically meaningful jet properties in compact scalar form and are used as inputs to dense neural network branches; the third one consists of 100×100 calorimeter images from the electromagnetic calorimeter (ECAL) and hadronic calorimeter (HCAL), which provide additional information on the energy deposition pattern of each jet.

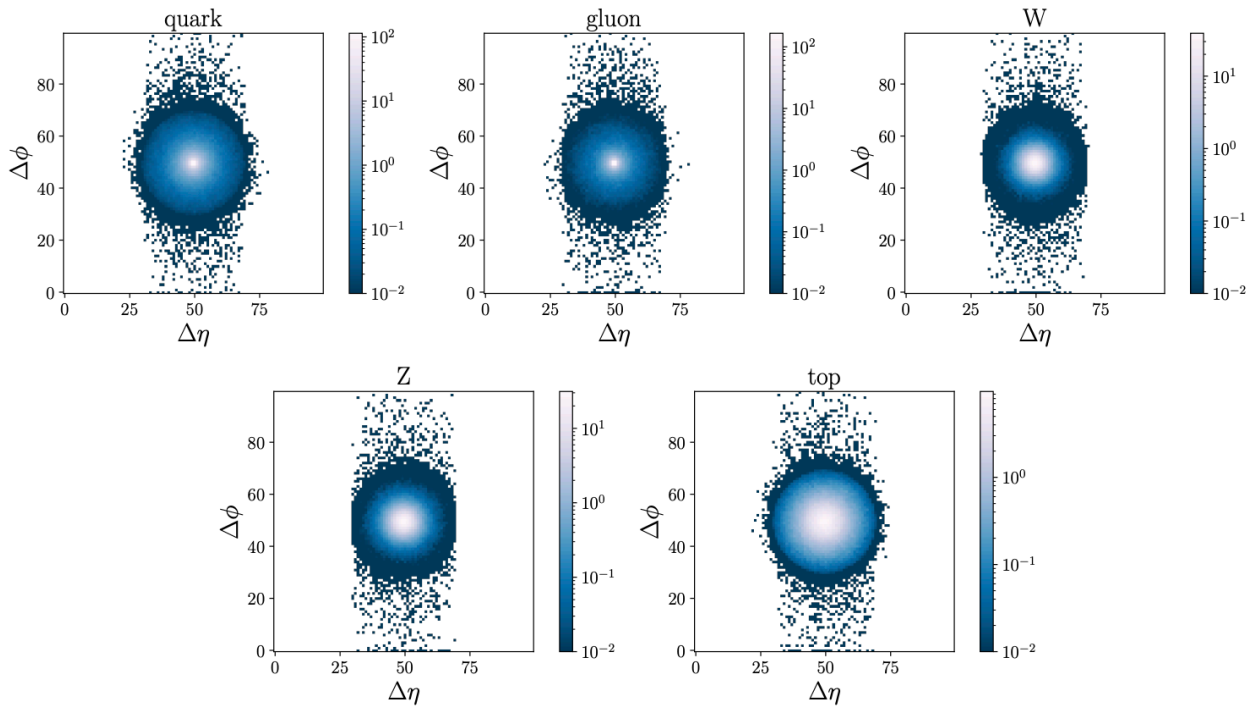

Figure 2: *Class-averaged 100 × 100 jet image representations for quark, gluon, W, Z and top jets in the $(\Delta\eta, \Delta\phi)$ plane. Pixel intensity denotes the average transverse-momentum content accumulated in each cell, highlighting the characteristic spatial structure of each jet class.*

Before training, several preprocessing steps were applied to the raw data. Firstly, the image inputs were cropped from 100×100 to 40×40 pixels in order to reduce the computational cost while retaining the central region containing the most relevant jet information; secondly the high-level features were standardised using `StandardScaler`, which improves numerical stability during optimisation and allows the network to learn from inputs defined on different physical scales. In addition, a mismatch was observed between the number of ECAL images and the corresponding jet and HCAL images (due to the presence of many nearly empty ECAL events); thus, in order to ensure consistent input dimensions across the multimodal models, ECAL images were duplicated where necessary to match the sample size of the other image arrays. These preprocessing steps produced a consistent set of aligned image and feature inputs suitable for training the different neural network architectures considered in this study.

2.2 Training

All classification models were trained on datasets obtained from eight `.h5` files containing simulated LHC jet events, for a total of 80,000 images. The jet images were centrally cropped from 100×100 to 40×40 pixels, as said before, and reshaped to $40 \times 40 \times 1$ for CNN input; subsequently the data were split into training, validation and test sets in a 70%/15%/15% ratio using a fixed random seed of 69. Training was carried out using the Adam optimiser, with a standard learning rate of 0.001, and in the final implementation, each model was trained for 30 epochs with the default Keras batch size used unless otherwise specified. For models including scalar jet observables, the same train-validation-test partitioning was applied to each feature input, and validation accuracy and loss were tracked throughout training to assess convergence.

3 Models Architecture

3.1 Jet Image CNN

The first model considered is a standard convolutional neural network trained using only the jet image representation. The image branch takes a single $40 \times 40 \times 1$ input and applies three convolutional layers with 32, 64 and 64 filters respectively, each followed by 2×2 max-pooling; a dropout layer is then used before flattening the output into a one-dimensional latent representation of size 576.
This flattened representation is passed through a sequence of dense layers with 256, 128, 128, 64, 64, 48, 32 and 16 neurons, using a mixture of ReLU and sigmoid activation functions, before a final 5-node softmax output layer classifies the jet as gluon, quark, W, Z or top. This model provides the baseline against which the more complex multimodal architectures are compared and its architecture is described in Fig. 3.

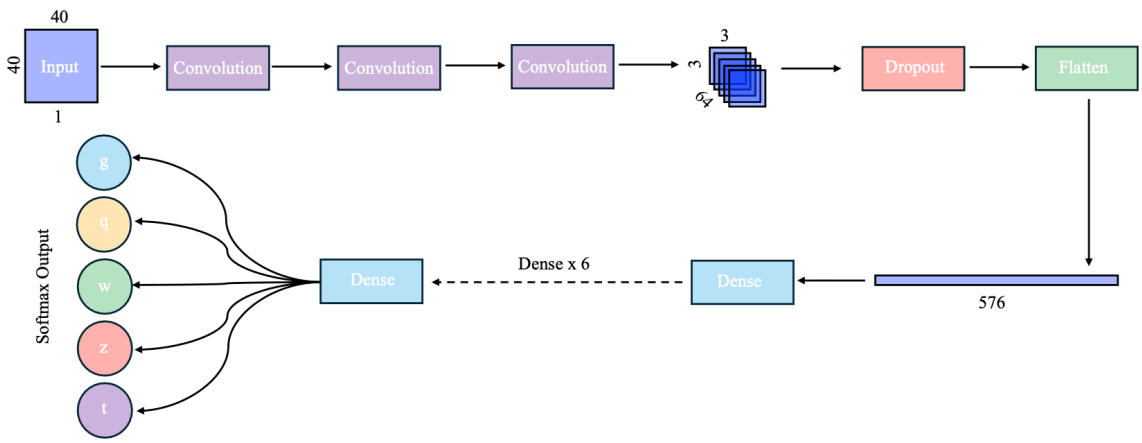

Figure 3: *Schematic representation of the neural network architecture used in this study*

3.2 Jet image + Features CNN-MLP

The second model combines the jet image representation with a set of 16 scalar jet features in a multimodal CNN–MLP architecture: a Multilayer Perceptron (MLP) is a class of feed-forward neural network composed of fully connected layers, in which each neuron is connected to every neuron in the subsequent layer through weighted links; unlike CNNs, which rely on convolutional filters to extract spatial patterns from image data, MLPs are particularly well suited to handling tabular inputs and

can learn non-linear decision boundaries. The purpose of this model is to test whether the spatial information extracted from the jet image can be improved by supplementing it with high-level kinematic and substructure observables.

The image branch follows the same structure as the baseline CNN, taking a cropped $40 \times 40 \times 1$ jet image and processing it through three convolutional layers with 32, 64 and 64 filters, each followed by max-pooling, dropout and flattening. In parallel, the scalar branch takes 16 jet features as separate inputs, including transverse momentum (p_T), pseudorapidity (η), jet mass, multiplicity, jet splitting $z \log z$, and other jet shape variables $\tau_3^{\beta=1}$, $d_2^{\beta=2}$, $d_2^{\alpha=1,\beta=2}$, $n_2^{\beta=2}$, $n_2^{\beta=1}$, $\tau_{32}^{\beta=1}$, $c_1^{\beta=1}$, $c_2^{\beta=1}$, trimmed mass, mMDT mass and soft-drop mass. Each scalar input is first processed through its own dense sub-network with 128 and 64 neurons and ReLU activation functions; after that all scalar representations are concatenated with the flattened image representation, and the merged vector is passed through additional dense layers before a final 5-node softmax output layer performs the classification (as shown in Fig. 4).

This architecture can be interpreted as a natural extension of the baseline CNN, since it combines low-level information, learned directly from the jet image, with physics observables that encode complementary information about jet substructure.

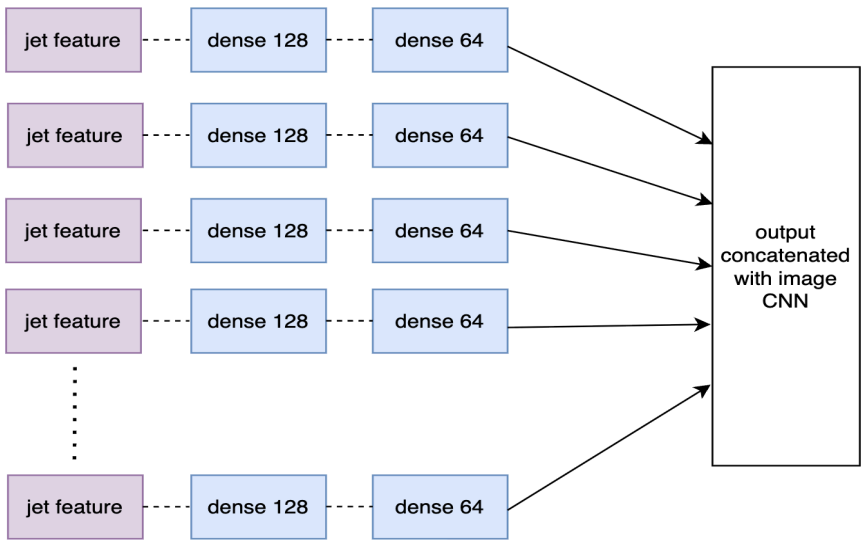

Figure 4: *Schematic representation of the scalar jet features branch*

3.3 Jet image + ECAL + HCAL Multiview CNN

The third model is a multi-view convolutional neural network that takes three image-based representations of the jet as input: the combined jet image, the ECAL image and the HCAL image.

Each of the three image branches receives a cropped 40×40 image and applies the same convolutional pipeline used in the baseline CNN, consisting of three convolutional layers with 32, 64 and 64 filters, max-pooling after each convolution, dropout, and flattening; the three flattened outputs are then concatenated into a combined latent vector of size three times larger than that of the single-image CNN. This merged representation is passed through a sequence of 6 dense layers and finally through a 5-node softmax output layer to classify the jet into the five particle categories, as shown in Fig. 5.

By incorporating the jet image together with its ECAL and HCAL calorimeter views, this model tests whether a multi-view treatment of detector information improves discrimination between similar jet

classes.

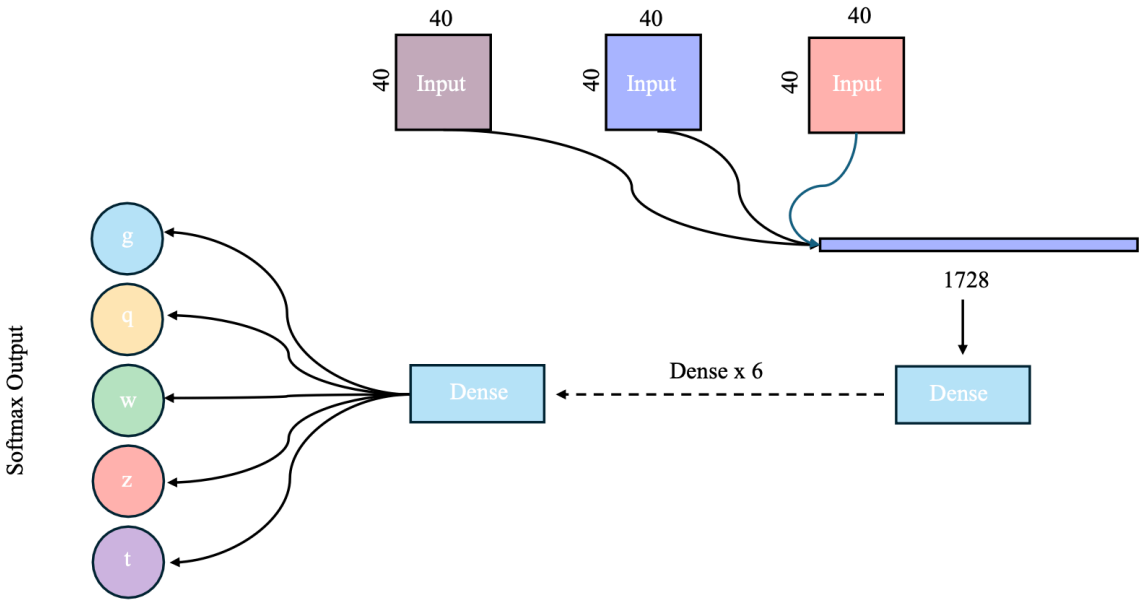

Figure 5: *Schematic representation of the neural network architecture used in this study*

3.4 Features only MLP

The fourth model uses only the 16 scalar jet features and does not include any image information: this architecture is a multilayer perceptron designed to test how much discriminative power is contained in the engineered kinematic and substructure observables alone.
The model takes the same 16 scalar jet features used in the CNN–MLP architecture as input, where each feature is treated as an individual scalar input and processed through dense layers; all branches are then concatenated and passed through further fully connected layers, with the aim of learning a joint representation of the high-level feature space. Finally a 5-node softmax output layer performs the classification into gluon, quark, W, Z and top jets (hence the schematic representation is the same as Fig. 4) .
This model provides an important benchmark because it isolates the predictive contribution of the scalar observables from that of the image-based detector representations.

3.5 Jet image + ECAL + HCAL + Features Multi-view CNN-MLP

The fifth and most complex model combines all available detector and feature information in a single multimodal architecture: it takes as input the jet image, the ECAL image, the HCAL image and the full set of 16 scalar jet features, with the aim of maximising classification performance by exploiting both low-level image structure and high-level engineered observables.
The three image branches follow the same convolutional structure as in the multi-view CNN, while the scalar inputs are processed through individual dense sub-networks as in the CNN–MLP model; the flattened image representations and all scalar feature representations are then concatenated into a single latent vector, which is passed through additional dense layers. The final layer is a 5-node softmax classifier that outputs the predicted jet class (its schematic representation is basically the union of architectures in Fig. 3 - Fig. 4 - Fig. 5).
This model represents the most information-rich architecture in the study and is intended to test the

maximum performance achievable when all detector views and scalar jet descriptors are used together.

4 Results

4.1 Performance of the models

Accuracy and loss are standard metrics for monitoring the behaviour of deep learning classifiers, providing a global summary of prediction quality and fit to the data [8]; however accuracy alone can be misleading when some classes are harder to identify than others, so we also report the average F1 score over the five jet classes, which combines precision and recall into a single, more balanced measure of classification performance [9].

Furthermore for multiclass classification problems, such as jet tagging across five particle types, confusion matrices provide an effective visualisation of the model's ability to discriminate between different classes [10]. In this analysis, we construct 5×5 confusion matrices where the main diagonal elements represent the true positive rate (correctly classified events) for each particle type, whereas the off-diagonal elements M_{ij} $(i \neq j)$ quantify the proportion of events belonging to class i that are misclassified as class j, revealing which particle types are most commonly confused with each other by the model.

Finally, the number of trainable parameters and the average epoch time characterise the computational cost of each model, allowing us to assess the trade-off between physics performance and resource usage:

- **Jet Image CNN**
  The CNN achieves satisfactory accuracy of 73%. From the confusion matrix, many gluons and quarks exhibit the highest mutual confusion, reflecting their similar QCD radiation patterns, and many Z bosons are misidentified as W bosons as well. CNN performances are summarized in Fig. 6.

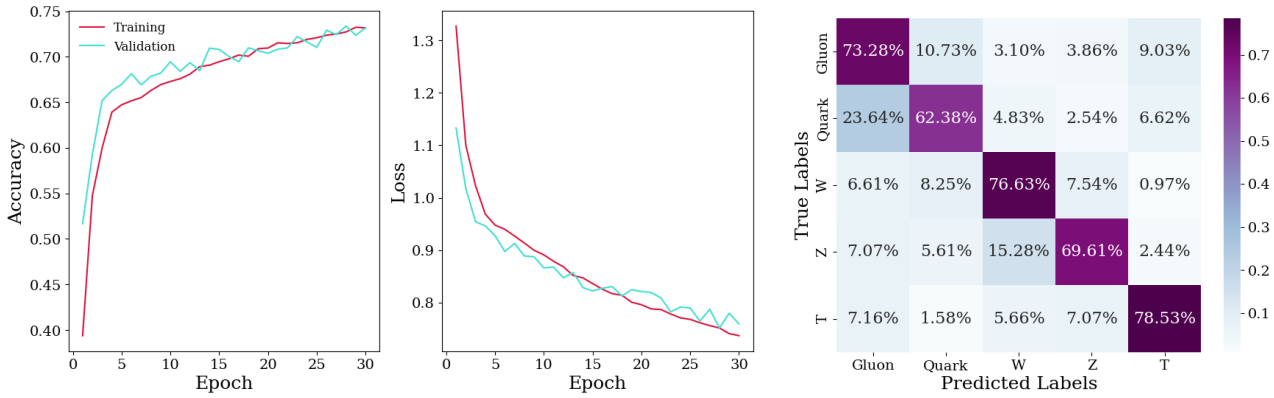

Figure 6: ***Left:*** *Accuracy vs number of epochs.* ***Middle:*** *Loss vs number of epochs.* ***Right:*** *Normalized confusion matrix.*

- **Jet Image + Features CNN–MLP**
  The CNN-MLP achieves excellent validation accuracy of 96.6%, with stable training curves (Fig. 7). The confusion matrix shows near-perfect separation: Top (99.8%) and Quark (99.9%) dominate, while Z bosons exhibit minimal confusion with W bosons (13.7%) due to their physical similarities.

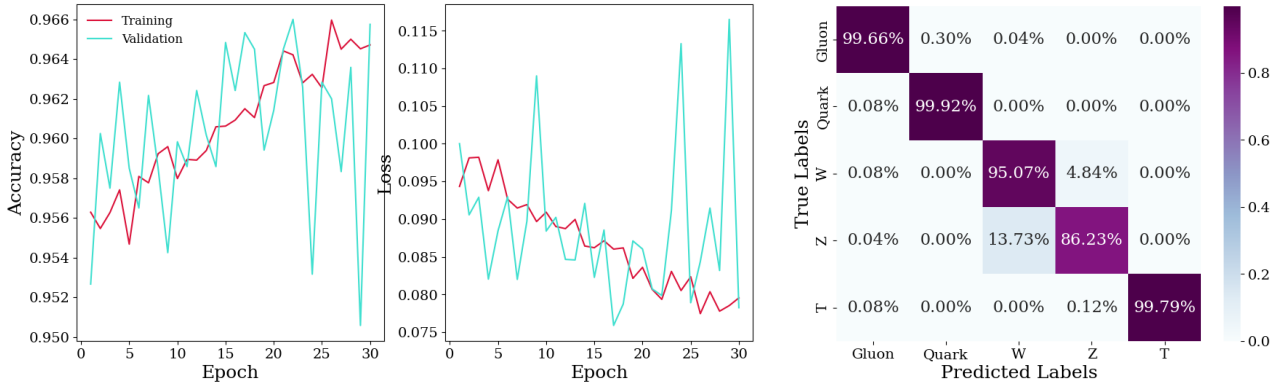

Figure 7: ***Left:*** *Accuracy vs number of epochs.* ***Middle:*** *Loss vs number of epochs.* ***Right:*** *Normalized confusion matrix.*

- **Jet image + ECAL + HCAL Multiview CNN**
  The Multiview CNN outperforms the baseline CNN achieving 91% validation accuracy and significantly lower loss of 22% , as shown in Fig. 8, thanks to the increased model complexity from three image inputs (401k parameters vs CNN's 270k). The normalized confusion matrix shows substantially fewer misidentifications, with Gluon/W/Z $>$ 90% and reduced overall particle confusion; thus the additional ECAL/HCAL views provide complementary calorimeter information which enable better jet discrimination.

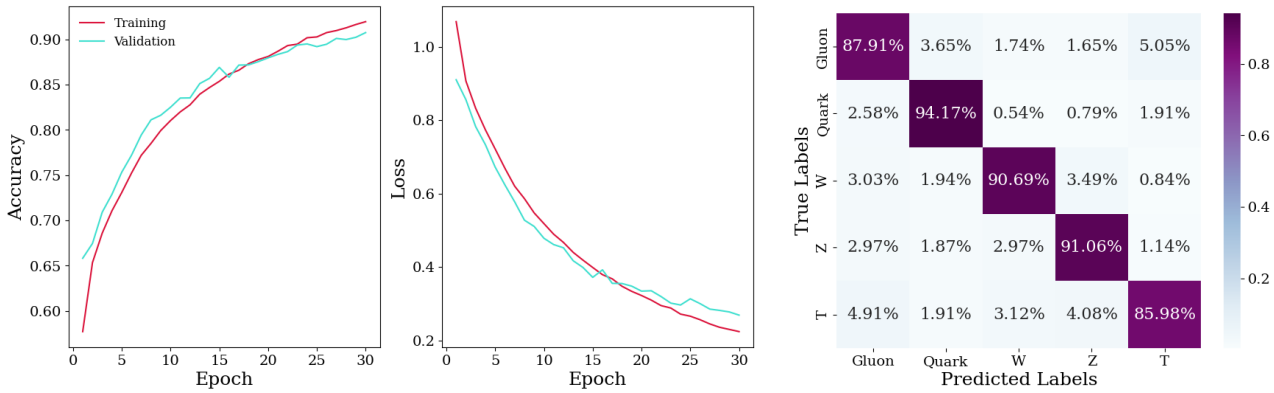

Figure 8: ***Left:*** *Accuracy vs number of epochs.* ***Middle:*** *Loss vs number of epochs.* ***Right:*** *Normalized confusion matrix.*

- **Features only MLP**
  By using only 16 High-Level Features (HLFs) without images, the MLP trains fastest while achieving satisfactory 91.8% validation accuracy (Fig. 9); overfitting is minimal with stable curves and low final loss. The confusion matrix shows strong Gluon/Quark (96-98%) and Top (99.8%) performance, even though W/Z retain relevant mutual confusion (13% and 26% respectively).

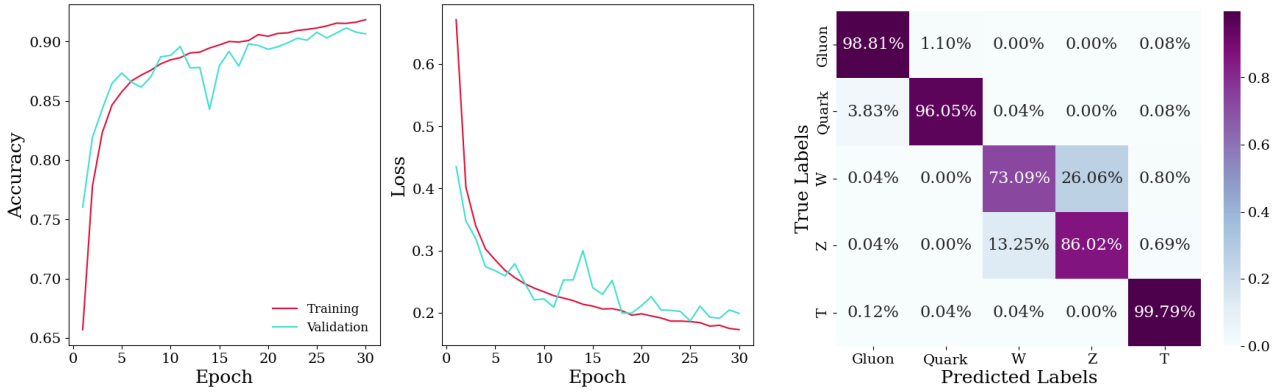

Figure 9: ***Left:*** *Accuracy vs number of epochs.* ***Middle:*** *Loss vs number of epochs.* ***Right:*** *Normalized confusion matrix.*

- **Jet image + ECAL + HCAL + Features Multi-view CNN-MLP**
  The final multimodal model integrating Jet/ECAL/HCAL images with 16 HLFs is unsurprisingly the most accurate one, with a 98% validation accuracy and minimal loss. The confusion matrix confirms excellent separation since Top/Quark/Gluon $\sim$ 99%, and W/Z present reduced confusion ($\sim$5%); as summarised in Fig. 10 the model outperforms all the previous ones, with the disadvantage of needing the longest time to be trained.

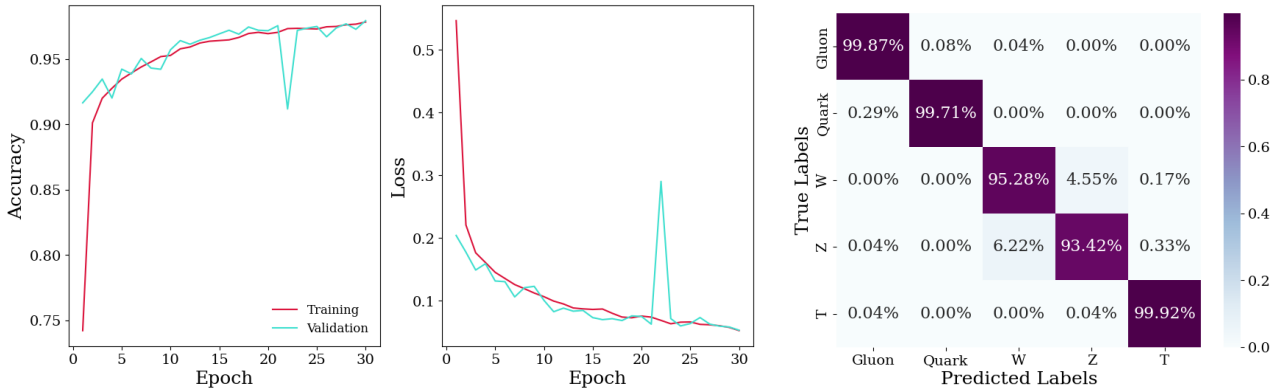

Figure 10: ***Left:*** *Accuracy vs number of epochs.* ***Middle:*** *Loss vs number of epochs.* ***Right:*** *Normalized confusion matrix.*

To compare the different architectures in a consistent way, we evaluate every single model using test accuracy, loss, average F1 score, number of trainable parameters and average training time per epoch, as shown in Tab. 1.

Model	**N. of Params**	**Epoch Time [s]**	**Loss**	**Accuracy**	**Avg F_1**
Images	270,581	16	0.737	0.732	0.721
Images + 16 HLFs	663,941	24	0.080	0.966	0.961
16 HLFs	460,741	5	0.173	0.919	0.907
Images + CAL Images	401,285	48	0.224	0.920	0.900
All Inputs	1,136,133	54	0.053	0.978	0.976

Table 1: *Performance metrics for different jet classification architectures. The model combining jet images with 16 HLFs offers the optimal performance-time tradeoff, achieving 96.6% accuracy with only 24s per epoch.*

4.2 Which particle classes are easiest to distinguish

The Receiver Operating Characteristic (ROC) curve plots the true positive rate against the false positive rate at varying classification thresholds, while the Area Under the Curve (AUC) quantifies the model's discriminative ability, with values near 1 indicating excellent separability [11]. From the comparison of Fig. 11 it can be clearly seen that the class most easily recognized (hence, whose curves are closer to the top left corner, and whose average AUC is closer to 1) is Top.

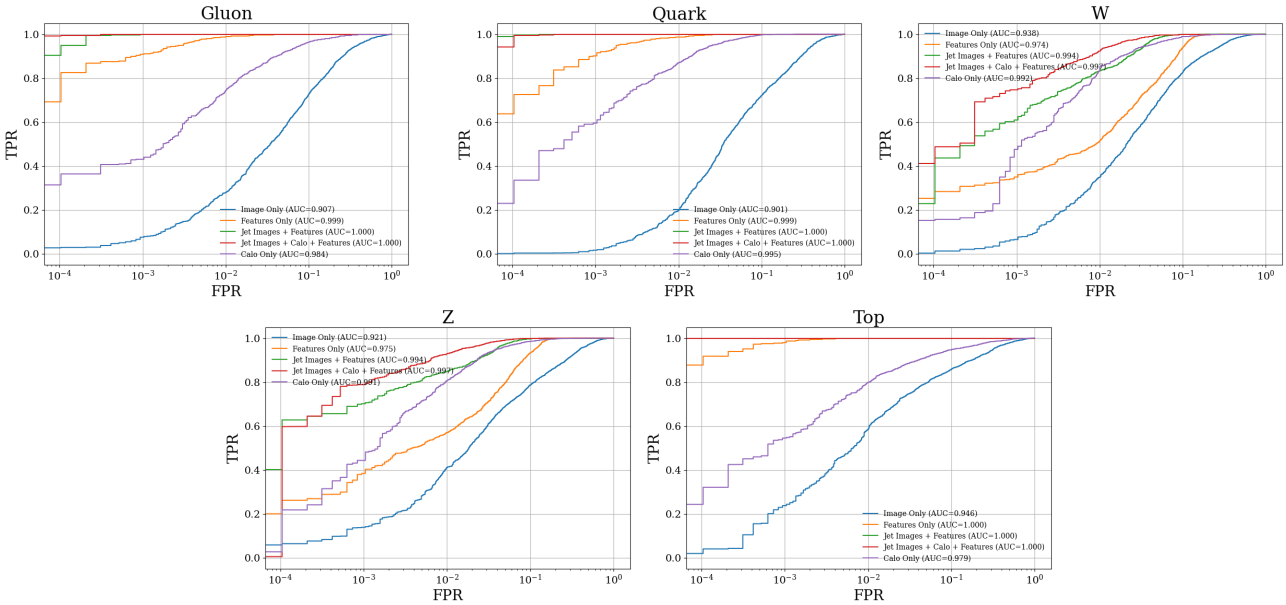

Figure 11: *Multi-class ROC curves for LHC jet classification*

4.3 Impact of individual features on performance

If we just focus on the 6 features the base CNN model was initially trained with (one at the time), which are transverse momentum (p_T), pseudorapidity (η), jet mass, multiplicity, jet splitting ($z \log z$) and $\tau_3^{\beta=1}$, we get the results in tab. 2 and Fig. 12.

Feature	**Avg Epoch Time [s]**	**Loss**	**Accuracy**	**Avg F_1**
Images	16	0.737	0.732	0.72
p_T	17	0.638	0.766	0.76
η	19	0.692	0.747	0.73
Mass	21	0.319	0.871	0.87
Multiplicity	20	0.397	0.850	0.82
$z \log z$	20	0.480	0.824	0.79
$\tau_3^{\beta=1}$	21	0.374	0.859	0.83

Table 2: *Performance metrics for CNN models trained on single jet features.*

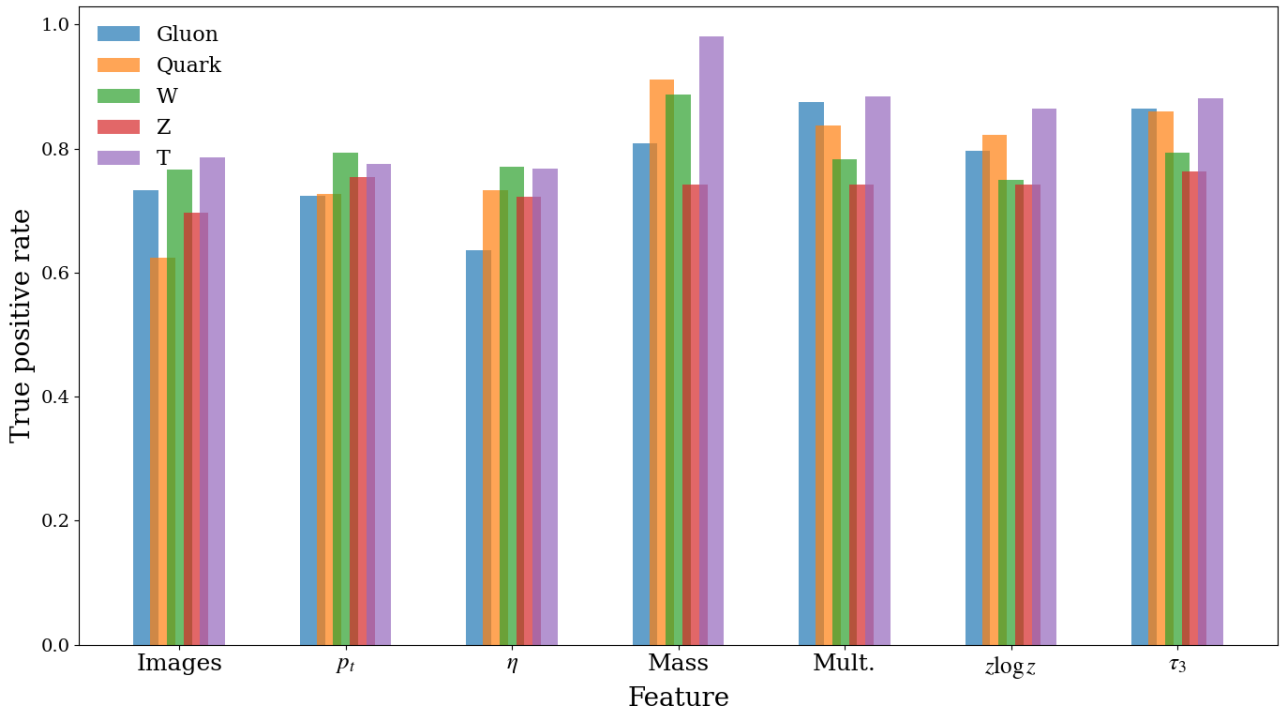

Figure 12: *True positive rate for each particle class (gluon, quark, W, Z, top) obtained using different input features. Mass and multiplicity yield the highest per-class TPRs, while η provides slightly lower performance.*

5 Conclusion

Five architectures were systematically evaluated for LHC jet classification, using various image and feature combinations, and achieving consistently strong performance for multi-input models (accuracy $> 90\%$). The most accurate configuration, combining jet/ECAL/HCAL images with 16 HLFs, delivered 97.8% accuracy and 0.976 F_1 score, though at highest computational cost. On the other hand, the model combining jet images with 16 HLFs offered similar performance (96.6% accuracy) with substantially faster training (24 s/epoch), representing the optimal accuracy-efficiency tradeoff for practical deployment. The class-wise analysis also showed that the usefulness of additional inputs depends on the particle category, confirming that physically meaningful scalar features provide complementary information rather than a uniform improvement across all classes. Taken together, these results support the use of multimodal deep learning as a robust and flexible approach for jet classification in high-energy physics.

Even though challenges persist (e.g. model scalability and potential overfitting from redundant features), precise jet tagging is crucial for LHC precision measurements and searches beyond the Standard Model.

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